



Cite this: *Phys. Chem. Chem. Phys.*,
2016, 18, 23184

Dynamics of ionic liquids in bulk and in confinement by means of ^1H NMR relaxometry – BMIM-OcSO₄ in an SiO₂ matrix as an example

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^1H nuclear magnetic resonance relaxometry is applied to reveal information on the translational and rotational dynamics of the ionic liquid: 1-butyl-3-methyl imidazoliumoctyl sulfate (BMIM-OcSO₄) in bulk and in a confinement formed by a nanoporous SiO₂ matrix. The experimental studies were performed in a very broad frequency range, from 8 kHz to 40 MHz (referring to the ^1H resonance frequency), in order to probe motional processes at very different time scales using a single experiment, and in the temperature range of 243–303 K. The relaxation results for BMIM-OcSO₄ in bulk are interpreted in terms of three relaxation contributions: a term associated with the translational dynamics of the ions (it has been assumed that the translational dynamics of cations and anions can be described by one diffusion coefficient) and two terms associated with the rotational motion of the anion and the cation, respectively. The relationships between the obtained dynamic parameters (rotational correlation times and translational diffusion coefficients) are thoroughly discussed and used as a “reference” for the dynamics of BMIM-OcSO₄ confined in an SiO₂ matrix. Analysis of the corresponding relaxation data for the confined liquid shows that the confinement does not significantly affect the rotational dynamics, but it has a considerable impact on the translational motion. It is demonstrated that the relaxation term associated with the translational dynamics stems from two contributions: a contribution from a core (bulk-like) fraction of the liquid and from a fraction moving near the pore surface and therefore being for some time adsorbed on the pore walls. The translational diffusion coefficient for the last fraction is determined and several conclusions regarding the residence lifetime of the ions on the surface are drawn. Moreover, an additional motional process on the timescale of ns or shorter is revealed in the confinement.

Received 10th April 2016,
Accepted 22nd July 2016

DOI: 10.1039/c6cp02377k

www.rsc.org/pccp

1. Introduction

The mechanisms of molecular and ionic dynamics in soft and hard matter are some of the most important questions of fundamental and applied science. This subject is extensively investigated by means of theoretical, experimental and computational methods. The properties of liquids in confinement attract special attention. Nevertheless, in most cases, research is directed towards water^{1–3} (because of its relevance to biology) and polymers.^{4–6} The physical and chemical properties of confined molecules are very different than in bulk. Studies concerned with phase transition behavior show that geometrical restrictions lead to a shift of the glass transition temperature,

melting and freezing points;^{7–9} for instance water confined in nanotubes shows a liquid-like behavior at temperatures as low as about 50 K.⁷ Such studies are often accompanied by molecular dynamics simulations attempting to explain the observed phenomena at the molecular level.^{7,10}

In this work nuclear magnetic resonance (NMR) relaxometry is exploited in order to investigate the dynamic properties of an ionic liquid: 1-butyl-3-methyl imidazoliumoctyl sulfate (BMIM-OcSO₄) in bulk and in a confinement formed by an SiO₂ matrix. NMR relaxometry is a unique experimental method to probe the mechanisms and characteristic time constants of dynamical processes in condensed matter on the molecular (atomistic) level. Therefore this method has earned the name “molecular rheology”. The remarkably broad range of magnetic fields covered by NMR relaxometry: from about 10 kHz to 40 MHz (referring to the ^1H resonance frequency) implies that one can investigate, using a single experiment, motional processes across a huge range of time scales (from ms to ns). In this context one should stress that “classical” relaxation experiments are performed at a single frequency.

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NMR relaxation experiments provide a quantity referred to as the spin–lattice relaxation rate, R_1 , (*i.e.* a reciprocal spin–lattice relaxation time, $R_1 = T_1^{-1}$). The dependence of the relaxation rate on the resonance frequency, $R_1(\omega)$, is called a relaxation dispersion profile. The predominating origin of ^1H relaxation in liquids is provided by ^1H – ^1H magnetic dipole–dipole interactions that stochastically fluctuate in time due to molecular (ionic) motion. The motion is characterized by a time correlation function of a mathematical form depending on the motional mechanism. The Fourier transform of the correlation function is referred to as a spectral density function. According to spin relaxation theories the relaxation rate, $R_1(\omega)$, can be described as a linear combination of the spectral densities taken at frequencies determined by a quantum-mechanical scenario of the relaxation process.^{11–16} In this way the shapes of the relaxation dispersion profiles reveal the underlying mechanisms of motional processes (besides their characteristic time constants).

Recently NMR relaxometry has extensively been applied to investigate the translational and rotational dynamics of molecular liquids.^{6,17} In particular, it has been demonstrated that translational diffusion coefficients can be determined from the low frequency slope of the relaxation rate plotted *versus* the square root of the resonance frequency. For long times the correlation function describing translational dynamics follows the power law: $C(t) \propto t^{-3/2}$ that is characteristic of free diffusion. This implies the form of the low frequency expansion of the translational spectral density, $J_{\text{trans}}(\omega) = a - b\sqrt{\omega}$,^{18,19} where b includes the translational diffusion coefficient. The power of this method lies in its simplicity, but it has only been exploited recently.²⁰ This approach has also been applied to ionic liquids in bulk.²¹ As ionic liquids contain different kinds of NMR active nuclei (they are, for instance, composed of ^1H containing cations and ^{19}F containing anions) the description has been extended taking into account all the relevant relaxation pathways (^1H – ^1H , ^1H – ^{19}F , ^{19}F – ^{19}F).²¹

Encouraged by those studies which have given access to the translational dynamics of both species, cations and anions, in a single experiment, we have decided to use NMR relaxometry to reveal information on the translational and rotational dynamics of BMIM–O cSO_4 in bulk and in SiO $_2$ confinement. It is worth mentioning that the use of ionic liquids for electrochemical applications is limited by their liquid nature – therefore they are inserted into porous gels (often SiO $_2$).

This work has two intertwined goals. The first one is to exploit the unique opportunity of NMR relaxometry to experimentally reveal (on the ionic level) the influence of confinement on the ionic dynamics. Experimental studies of the molecular (ionic) dynamics of confined liquids are rare. In most cases only the macroscopic properties of confined liquids (like phase transition temperatures) are investigated and measured. Such studies are accompanied by molecular dynamics simulations in order to explain on the molecular (ionic) level the observed changes in the macroscopic quantities caused by confinement. Investigating confined systems means gaining insights into the specifics of motion in restricted geometry depending on the properties of the confinement and the liquid. One can expect

that the dynamic properties of ionic liquids in confinement strongly depend on the relative size of the ions and the pores. Most ionic liquids are relatively large molecules which are expected to be significantly affected by the geometrical restrictions of nano-sized confinements. The expectations are confirmed by studies of the phase transition behavior of ionic liquids in confinement which reveal that indeed the melting point, crystallization temperature and glass transition temperature significantly change as a result of the confinement conditions.^{8,9} To enhance the effects a liquid containing large anions (O cSO_4) has been chosen. The trajectories of molecules near the surface can be described as a succession of adsorption steps and flights in the ‘free’ fraction. One can distinguish here between the weak adsorption and strong adsorption cases.²² In the weak adsorption case molecules return to the ‘free’ fraction after a single adsorption act, while in the strong adsorption case the molecules frequently return to the surface before they indeed escape to the ‘free’ fraction.^{22,23} The statistical description of Brownian motion near a surface^{24,25} combined with spin relaxation theory leads to characteristic power laws of the relaxation rate *versus* frequency^{22,23} that allow us to distinguish between these cases. This very interesting concept has been used to discuss the dynamics of water in LAPONITE $^{\text{®}}$.^{22,26} Nevertheless as the power factor depends on several quantities (like the dipolar interaction strength during the adsorption step, the characteristic time constants of adsorption and diffusion, a parameter describing the shape of the surface) that can hardly be independently determined, the information obtained in this way is rather limited. In this work we apply a different theoretical approach, based on a detailed comparison of the ^1H spin–lattice relaxation dispersion profiles for the bulk and confined liquid, with the aim to explicitly determine the dynamic parameters (translational diffusion coefficients, rotational correlation times) for both cases.

The second goal of this work is to explore the potential (and limitations) of NMR relaxometry as a method of investigating the dynamics of complex systems. Ionic liquids can be definitely considered as such ones. The geometrical restrictions and interactions with the surface of confining walls are sensed by a network of interacting ions, affecting the dynamics of both species (cations and anions). This binary nature of the liquid makes the analysis of the relaxation data much more demanding than for simple molecular liquids.

The paper is organized as follows: in Section 2 the theory of ^1H spin–lattice relaxation is presented, Section 3 contains experimental details, while a thorough analysis of the relaxation data for BMIM–O cSO_4 in bulk and in SiO $_2$ confinement is presented in Section 4. The results are discussed in Section 5 and eventually concluded in Section 6.

2. Theory

^1H relaxation in liquids is predominantly caused by magnetic dipole–dipole interactions between protons. As the dipolar interactions can be of intramolecular and intermolecular origin,

the overall spin–lattice relaxation rate, $R_1(\omega)$ (ω denotes the ^1H resonance frequency in angular frequency units), is given as a sum of the corresponding contributions:^{6,17}

$$R_1(\omega) = R_{1,\text{intra}}(\omega) + R_{1,\text{inter}}(\omega) \quad (1)$$

where $R_{1,\text{intra}}(\omega)$ and $R_{1,\text{inter}}(\omega)$ are the intramolecular and intermolecular contributions, respectively. The intramolecular interactions fluctuate in time due to molecular (ionic) rotation, while the intermolecular dipolar couplings are mainly modulated by the relative translational diffusion of the interacting species. As a consequence, the relaxation dispersion profiles give access to both the rotational and the translational dynamics. As in the case of BMIM-OcSO₄ both ions (the cation and the anion) contain protons, the relaxation rate $R_1(\omega)$ includes five relaxation channels corresponding to three intermolecular couplings: cation–cation (C–C), cation–anion (C–A) and anion–anion (A–A) and two intramolecular interactions: for the cation (C) and for the anion (A):

$$R_1(\omega) = R_{1,\text{inter}}^{\text{C–C}}(\omega) + R_{1,\text{inter}}^{\text{C–A}}(\omega) + R_{1,\text{inter}}^{\text{A–A}}(\omega) + R_{1,\text{intra}}^{\text{C}}(\omega) + R_{1,\text{intra}}^{\text{A}}(\omega) \quad (2)$$

The intramolecular relaxation contributions, $R_1^{\text{I}}(\omega)$ (I = A, C), are given as a linear combination of spectral density functions, $J_{\text{rot}}^{\text{I}}$:

$$R_1^{\text{I}}(\omega) = C_{\text{DD}}^{\text{I}}(J_{\text{rot}}^{\text{I}}(\omega) + 4J_{\text{rot}}^{\text{I}}(2\omega)) \quad (3)$$

where C_{DD}^{I} denotes a dipolar relaxation constant for the cation (I = C) and the anion (I = A). The spectral densities are Fourier transforms of the corresponding time correlation functions. In the simplest case of free isotropic rotation the spectral densities assume a Lorentzian form (corresponding to exponential correlation functions):

$$J_{\text{rot}}^{\text{I}}(\omega) = \frac{\tau_{\text{rot}}^{\text{I}}}{1 + (\omega\tau_{\text{rot}}^{\text{I}})^2} \quad (4)$$

where $\tau_{\text{rot}}^{\text{I}}$ denotes the rotational correlation times for the cation and for the anion, I = C and I = A, respectively. It has been demonstrated²⁷ that due to different mathematical forms of the spectral densities characterizing rotational and translational motions and the time scale separation of both processes, the relaxation profiles can unambiguously be separated into contributions associated with rotational and translational dynamics. As far as the time scale separation is concerned, according to the Stokes–Einstein equation for spherical particles, the ratio between translational (τ_{trans}) and rank-two rotational (τ_{rot}) correlation times yields: $\tau_{\text{trans}}/\tau_{\text{rot}} = 9^{11}$; it has been shown that for “real” molecular and ionic liquids the ratio is typically of the order of 30–40.^{17,28} On the other hand it has been demonstrated that the translational diffusion coefficients of cations and anions in ionic liquids are similar.²¹ As a consequence, one can hardly expect that the three translational contributions (C–C, C–A, A–A) of eqn (2) can be separately revealed from the relaxation dispersion profile. Being aware of this fact, let us assume that the translational diffusion coefficients for both ions are equal: $D^{\text{C}} = D^{\text{A}} = D$. Under this

assumption the intermolecular contribution to the overall relaxation can be described as:

$$\begin{aligned} R_{1,\text{inter}}(\omega) &= R_{1,\text{inter}}^{\text{C–C}}(\omega) + R_{1,\text{inter}}^{\text{C–A}}(\omega) + R_{1,\text{inter}}^{\text{A–A}}(\omega) \\ &= \frac{3}{10} \left(\frac{\mu_0}{4\pi} \gamma_{\text{H}}^2 \hbar \right)^2 \frac{N_{\text{H}}}{d^3} (J_{\text{trans}}(\omega) + 4J_{\text{trans}}(2\omega)) \end{aligned} \quad (5)$$

where N_{H} is the number of protons per unit volume (not distinguishing between the cation and anion protons), while d denotes the distance of closest approach of ions (again, we do not distinguish between cations and anions); γ_{H} is the proton gyromagnetic factor, μ_0 and $\hbar$ denote the vacuum permeability and the Planck constant, respectively. Employing the force free hard sphere model of translational diffusion,^{29,30} according to which molecules (ions) are treated as hard spheres uniformly distributed outside the distance of closest approach (which is equal to a sum of the radii of the interacting molecules), the translational spectral density is given as:

$$J_{\text{trans}}(\omega) = 72 \int_0^\infty \frac{u^2}{81 + 9u^2 - 2u^4 + u^6} \frac{u^2}{u^4 + (\omega\tau_{\text{trans}})^2} du \quad (6)$$

The translational correlation time, τ_{trans} , is defined as: $\tau_{\text{trans}} = d^2/D_{12}$, where D_{12} denotes a relative translational diffusion coefficient defined as a sum of the diffusion coefficients of the interacting species; under the assumption that $D^{\text{C}} = D^{\text{A}} = D$, one gets: $D_{12} = 2D$. As a result, the relaxation dispersion profiles, $R_1(\omega)$, presented in Section 4 are analyzed in terms of the explicitly written expression:

$$\begin{aligned} R_1(\omega) &= R_{1,\text{inter}}(\omega) + R_{1,\text{intra}}^{\text{C}} + R_{1,\text{intra}}^{\text{A}} = \frac{108}{5} \left(\frac{\mu_0}{4\pi} \gamma_{\text{H}}^2 \hbar \right)^2 \frac{N_{\text{H}}}{d^3} \\ &\times \int_0^\infty \frac{u^2}{81 + 9u^2 - 2u^4 + u^6} \left(\frac{u^2}{u^4 + (\omega\tau_{\text{trans}})^2} + 4 \frac{u^2}{u^4 + (2\omega\tau_{\text{trans}})^2} \right) du \\ &+ C_{\text{DD}}^{\text{C}} \left(\frac{\tau_{\text{rot}}^{\text{C}}}{1 + (\omega\tau_{\text{rot}}^{\text{C}})^2} + 4 \frac{\tau_{\text{rot}}^{\text{C}}}{1 + (2\omega\tau_{\text{rot}}^{\text{C}})^2} \right) \\ &+ C_{\text{DD}}^{\text{A}} \left(\frac{\tau_{\text{rot}}^{\text{A}}}{1 + (\omega\tau_{\text{rot}}^{\text{A}})^2} + 4 \frac{\tau_{\text{rot}}^{\text{A}}}{1 + (2\omega\tau_{\text{rot}}^{\text{A}})^2} \right) \end{aligned} \quad (7)$$

The translational diffusion coefficient can also be determined from the low frequency slope of the relaxation rate plotted *versus* the square root of the frequency.^{18–20} At low frequencies, when $\omega\tau_{\text{trans}} < 1$, the relaxation rate $R_1(\omega)$ can be approximated by a linear function: $R_1(\omega) \cong R_1(0) - B\sqrt{\omega}$. The coefficient B includes the translational diffusion coefficient:^{18,19}

$$B = \pi N_{\text{H}} \left(\frac{\mu_0}{4\pi} \gamma_{\text{H}}^2 \hbar \right)^2 \left(\frac{1 + 4\sqrt{2}}{30} \right) D^{-3/2} \quad (8)$$

It is important to notice that the coefficient B includes, besides the diffusion coefficient and the number of protons per unit volume, only well-defined physical constants (it is independent of the distance of closest approach, d); the N_{H} value can be easily determined from the molar mass and density of the liquid. In addition to providing the value of the translational diffusion coefficient, the linear dependence $R_1(\omega) \cong R_1(0) - B\sqrt{\omega}$ observed at low frequencies can be treated as proof that the molecule (ion) undergoes three-dimensional diffusion. The method described

by eqn (8) has been used to determine the translational diffusion coefficients for several molecular liquids;²⁰ as far as ionic liquids are concerned it has been applied to EMIM-SCN.²¹ In this paper we shall exploit it for BMIM-OcSO₄.

3. Experimental details

Ionic liquid BMIM-OcSO₄ contained in a silica matrix was synthesized using a non-hydrolytic sol-gel route. The concentration of BMIM-OcSO₄ was 40 wt%. Details of confining the IL using a non-hydrolytic sol-gel route are given in ref. 31. Pore parameter analysis was performed at 77 K on a Gemini VII 2390t from Micromeritics Instrument Corporation. Before the nitrogen sorption measurements, the ionic liquid was extracted by washing and ultrasonating the sample in dichloromethane at 60 °C and then the washed sample was vacuum-dried for 12 h at 60 °C. The surface area was measured using the BET (Brunauer-Emmett-Teller) method, while the average pore diameter was determined using the Barrett-Joyner-Halenda (BJH) approach. The average pore size is 2.7 nm, while the surface area and pore volume are obtained as 356 m² g⁻¹ and 0.25 cm³ g⁻¹, respectively.

BMIM-OcSO₄ was purchased from Sigma-Aldrich. The structure of the compound is shown in Fig. 1. The relaxation data were collected using a STELAR Fast Field Cycling (FFC) relaxometer. The measurements were conducted for the following settings: the RF detection pulse length: 6.5 μs, the detection frequency: 15.8 MHz, the slew rate of the magnetic field: 12 MHz ms⁻¹, the repetition time was set to fivefold the spin-lattice time at the highest relaxation field – 40 MHz, the polarization time was set to fourfold the spin-lattice time at 24 MHz. The non-polarized sequence was used for fields above 10 MHz, while for lower fields pre-polarization at 24 MHz was applied. Temperature was controlled using a Eurotherm controller with an additional thermocouple temperature sensor inserted into the probehead. The temperature value was checked before and after measurements to ensure stability at the 0.5 K level. The range of ¹H resonance frequencies covered in the experiment is 8 kHz–40 MHz, while the temperature range yields: 243–303 K.

4. Results and analysis

4.1 BMIM-OcSO₄ in bulk

We begin the analysis with BMIM-OcSO₄ in bulk. ¹H spin-lattice relaxation dispersion profiles of BMIM-OcSO₄ are shown in Fig. 2 together with the corresponding theoretical fits in terms of eqn (7). The fitting strategy was as follows. The number of protons per unit

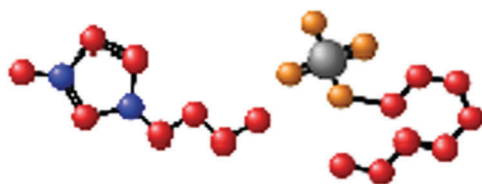


Fig. 1 Structure of BMIM cation (left) and OcSO₄ anion (right); C – red, N – blue, O – orange, S – gray.

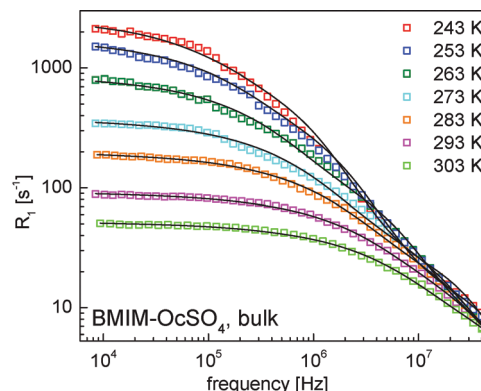


Fig. 2 ¹H spin-lattice relaxation dispersion profiles of BMIM-OcSO₄ in bulk. Solid lines – fits according to eqn (7).

value has been calculated from the expression: $N_H = \rho N_A n / M$, where $M = 348.5 \text{ g mol}^{-1}$ and $\rho = 1.06 \text{ g cm}^{-3}$ denote the molecular mass and density of the liquid, respectively, $n = 32$ is the number of protons in one molecule (including BMIM cations and OcSO₄ anions) and N_A is Avogadro's number; $N_H = 5.8 \times 10^{-2} \text{ Å}^{-3}$ has been obtained. Six adjustable parameters have been used: τ_{trans} and d to describe the translational motion, C_{DD}^{A} and $\tau_{\text{rot}}^{\text{A}}$ to characterize the anionic dynamics and C_{DD}^{C} and $\tau_{\text{rot}}^{\text{C}}$ to characterize the cationic dynamics. The parameters: d , C_{DD}^{A} and C_{DD}^{C} have been kept temperature independent. The values obtained from the analysis are summarized in Table 1 which also includes the translational diffusion coefficient calculated from the expression: $D = d^2 / (2\tau_{\text{trans}})$. As far as the temperature independent parameters are concerned: $d = 4.2 \text{ Å}$, $C_{\text{DD}}^{\text{A}} = 9.0 \times 10^8 \text{ Hz}^2$ and $C_{\text{DD}}^{\text{C}} = 1.47 \times 10^9 \text{ Hz}^2$ have been obtained. The ratio between the obtained rotational correlation times is about 20. As the OcSO₄ anion is considerably larger than the BMIM cation we attribute the longer correlation time (and the corresponding dipole-dipole relaxation constant) to the anionic dynamics. At higher temperatures, 273 K, 283 K and 293 K above, the rotational dynamics of the BMIM cations becomes too fast to allow for directly determining the correlation time. For 273 K, 283 K, 293 K and 303 K the contribution associated with the cationic rotation is represented as a frequency independent (constant) factor: $A = 5C_{\text{DD}}^{\text{C}}\tau_{\text{rot}}^{\text{C}}$ being the limit of the corresponding expression (eqn (7)) for $\omega\tau_{\text{rot}}^{\text{C}} \ll 1$. Thus, the correlation times have been determined from the A values setting $C_{\text{DD}}^{\text{C}} = 1.47 \times 10^9 \text{ Hz}^2$. At 303 K also the anionic rotation becomes too fast to lead to a significant relaxation dispersion. In this case both (cationic and anionic) relaxation contributions converge to a frequency independent term: $\tilde{A} = 5(C_{\text{DD}}^{\text{C}}\tau_{\text{rot}}^{\text{C}} + C_{\text{DD}}^{\text{A}}\tau_{\text{rot}}^{\text{A}})$ and hence at this temperature neither $\tau_{\text{rot}}^{\text{C}}$ nor $\tau_{\text{rot}}^{\text{A}}$ can be determined.

In Fig. 3 the overall relaxation dispersion profiles, $R_1(\omega)$, for the low temperatures (243 K, 253 K and 263 K) are decomposed into the individual contributions corresponding to the translational dynamics, rotational dynamics of anions and rotational dynamics of cations. An analogous decomposition for the higher temperatures (273 K, 283 K, 293 K and 303 K) is shown in Fig. 4. It should be reminded at this stage that the values of the translational diffusion coefficient presented in Table 1 have been obtained under the assumption that the translational dynamics of cations and anions is very similar (it has been set $D^{\text{C}} = D^{\text{A}} = D$).

Table 1 Parameters describing the translational and rotational dynamics of BMIM-OcSO₄ in bulk ($d = 4.2 \text{ \AA}$, $C_{\text{DD}}^{\text{A}} = 9 \times 10^8 \text{ Hz}^2$, $C_{\text{DD}}^{\text{C}} = 1.47 \times 10^9 \text{ Hz}^2$, $N_{\text{H}} = 5.8 \times 10^{-2} \text{ \AA}^{-3}$)

$T \text{ [K]}$	$\tau_{\text{trans}} \text{ [s]}$	$D \text{ [m}^2 \text{ s}^{-1}\text{]}$	$D^{(\text{slope})} \text{ [m}^2 \text{ s}^{-1}\text{]}$	$\tau_{\text{rot}}^{\text{A}} \text{ [s]}$	$\tau_{\text{rot}}^{\text{C}} \text{ [s]}$	Stand. dev. (%)
243	1.83×10^{-6}	4.82×10^{-14}	6.13×10^{-14}	8.49×10^{-8}	3.00×10^{-9}	12.2
253	1.26×10^{-6}	7.00×10^{-14}	6.81×10^{-14}	4.60×10^{-8}	1.83×10^{-9}	6.7
263	6.51×10^{-7}	1.36×10^{-13}	1.34×10^{-13}	1.48×10^{-8}	8.6×10^{-10a}	3.8
273	2.77×10^{-7}	3.18×10^{-13}	3.17×10^{-13}	8.01×10^{-9}	1.5×10^{-10a}	14.2
283	1.43×10^{-7}	6.17×10^{-13}	6.30×10^{-13}	5.26×10^{-9}	6.1×10^{-11a}	3.1
293	6.58×10^{-8}	1.34×10^{-12}	1.45×10^{-12}	2.06×10^{-9}	2.5×10^{-11a}	2.1
303	3.56×10^{-8}	2.48×10^{-12}	2.52×10^{-12}	—	—	7.6

^a $\tau_{\text{rot}}^{\text{C}}$ values obtained from the term $A = 5C_{\text{DD}}^{\text{C}}\tau_{\text{rot}}^{\text{C}}$ setting $C_{\text{DD}}^{\text{C}} = 1.47 \times 10^9 \text{ Hz}^2$. The A values are (in s^{-1}): 263 K: 6.31; 273 K: 3.66; 283 K: 1.49; 293 K: 0.61, while (for 303 K) $\dot{A} = 5(C_{\text{DD}}^{\text{C}}\tau_{\text{rot}}^{\text{C}} + C_{\text{DD}}^{\text{A}}\tau_{\text{rot}}^{\text{A}}) = 4.2 \text{ Hz}$.

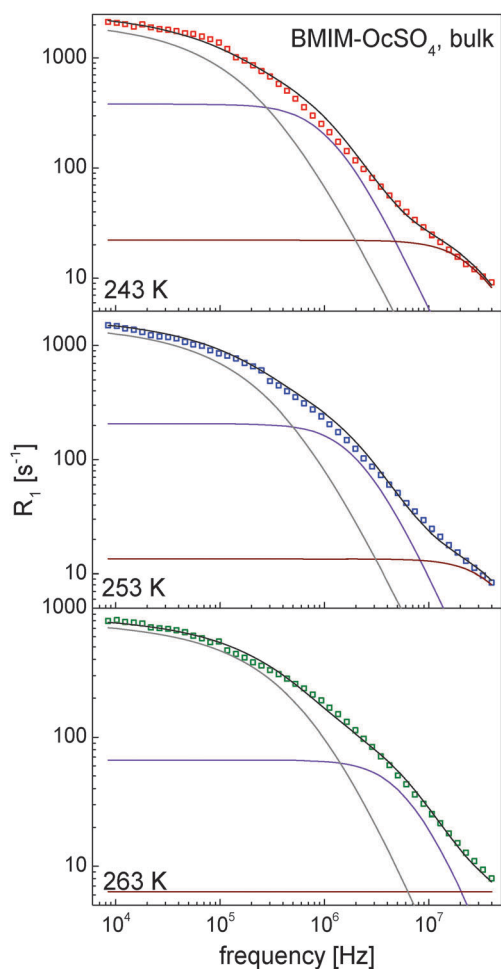


Fig. 3 ¹H spin-lattice relaxation profiles of BMIM-OcSO₄ in bulk at 243 K, 253 K and 263 K, decomposed into the contributions described by eqn (7): gray lines – inter-ionic (intermolecular) contribution ($R_{1,\text{inter}}(\omega)$), violet lines – anionic intramolecular contribution ($R_{1,\text{intra}}^{\text{A}}(\omega)$), brown lines – cationic intramolecular contribution ($R_{1,\text{intra}}^{\text{C}}(\omega)$), black lines – overall relaxation rate ($R_1(\omega)$).

The translational diffusion coefficient has also been determined from the low frequency slope of the relaxation rate $R_1(\omega)$ versus $\sqrt{\omega}$ (Fig. 5a and b) in terms of eqn (8). The values obtained from the full analysis and from this linear dependence are in very good agreement as shown in Fig. 6. Fig. 7 shows the correlation times, τ_{trans} , $\tau_{\text{rot}}^{\text{A}}$ and $\tau_{\text{rot}}^{\text{C}}$ versus reciprocal temperature.

In both figures (Fig. 6 and 7) the dynamical parameters obtained for BMIM-OcSO₄ are compared with the corresponding values for BMIM-OcSO₄ in the SiO₂ matrix determined in the section below.

4.2 BMIM-OcSO₄ in an SiO₂ matrix

The corresponding ¹H spin-lattice relaxation dispersion profiles for BMIM-OcSO₄ in an SiO₂ matrix are shown in Fig. 8. Two observations can be made. First, the relaxation data for BMIM-OcSO₄ in confinement show weaker variations with temperature than those for BMIM-OcSO₄ in bulk. Second, the relaxation dispersion profiles of BMIM-OcSO₄ in bulk and in confinement very significantly differ at higher temperatures; with decreasing temperature the differences are smaller, but still noticeable even at the lowest temperatures as shown in Fig. 9a and b for 243 K and 253 K. We begin the analysis with those temperatures. One can see in Fig. 9a that the relaxation rates for BMIM-OcSO₄ in the SiO₂ matrix at 243 K differ from the corresponding data for BMIM-OcSO₄ in bulk mainly at high frequencies (where the rotational dynamics gives the dominating contribution to the overall relaxation). At 253 K (Fig. 9b) the relaxation data for confinement and bulk start to differ also at low frequencies; the relaxation rates for the confined liquid are larger than for bulk, suggesting that also the translational dynamics (which dominates the relaxation at low frequencies) becomes slower in confinement. To quantitatively describe these effects the relaxation data for confined BMIM-OcSO₄ at 243 K and 253 K have been analyzed in terms of eqn (7) (in analogy to the case of BMIM-OcSO₄ in bulk) modified by adding a frequency independent term, C , which has turned out to be needed to reproduce the experimental data at high frequencies. The parameters: $\tau_{\text{trans}}^{\text{conf}}$, $\tau_{\text{rot}}^{\text{A,conf}}$, $\tau_{\text{rot}}^{\text{C,conf}}$ (where the index “conf” explicitly refers to the confinement) have been treated as adjustable. The other parameters: d^{conf} , $C_{\text{DD}}^{\text{A,conf}}$, $C_{\text{DD}}^{\text{C,conf}}$ have been set to the corresponding values obtained for BMIM-OcSO₄ in bulk, i.e.: $d^{\text{conf}} = d = 4.2 \text{ \AA}$, $C_{\text{DD}}^{\text{A,conf}} = C_{\text{DD}}^{\text{A}} = 9.0 \times 10^8 \text{ Hz}^2$, $C_{\text{DD}}^{\text{C,conf}} = C_{\text{DD}}^{\text{C}} = 1.47 \times 10^9 \text{ Hz}^2$; while $N_{\text{H}} = 5.8 \times 10^{-2} \text{ \AA}^{-3}$. The fits decomposed into the individual contributions are shown in Fig. 9a and b, while the obtained correlation times are collected in Table 2 (which also includes the calculated diffusion coefficient: $D^{\text{conf}} = d^2/(2\tau_{\text{trans}}^{\text{conf}})$) and shown, for comparison, in Fig. 6 (D^{conf}) and Fig. 7 ($\tau_{\text{trans}}^{\text{conf}}$, $\tau_{\text{rot}}^{\text{A,conf}}$, $\tau_{\text{rot}}^{\text{C,conf}}$). The values of these parameters and their relationships are discussed in Section 5.

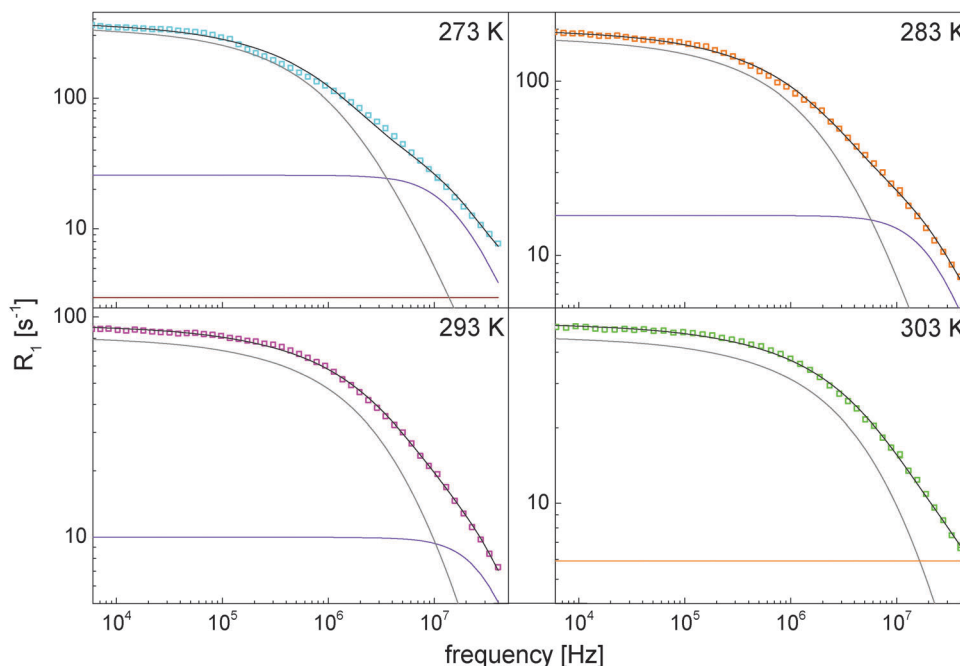


Fig. 4 ^1H spin-lattice relaxation profiles of BMIM-OcSO₄ in bulk at 273 K, 283 K, 293 K and 303 K, decomposed into the contributions described by eqn (7): gray lines – inter-ionic (intermolecular) contribution ($R_{1,\text{inter}}(\omega)$), violet lines – anionic intramolecular contribution ($R_{1,\text{intra}}^A(\omega)$), brown line (shown only for 273 K) – $A = 5C_{\text{DD}}^C\tau_{\text{rot}}$, orange line (303 K) – $\tilde{A} = 5(C_{\text{DD}}^C\tau_{\text{rot}} + C_{\text{DD}}^A\tau_{\text{rot}}) = 4.2$ Hz, black lines – overall relaxation rate ($R_1(\omega)$).

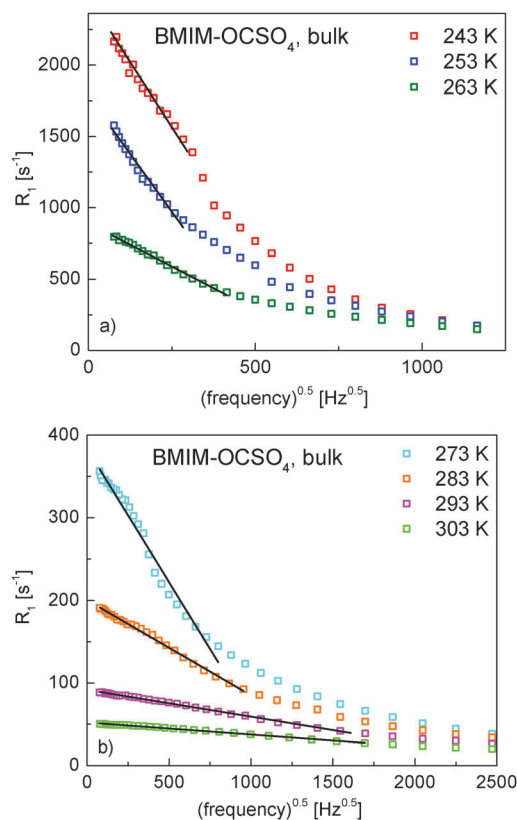


Fig. 5 (a and b) Linear dependencies of the ^1H spin-lattice relaxation rate, $R_1(\omega)$, of BMIM-OcSO₄ in bulk versus square root of frequency, $\sqrt{\omega}$, for (a) 243 K, 253 K, 263 K; (b) 273 K, 283 K, 293 K, 303 K; solid lines – linear fits. The B confident (eqn (8)) yields (in $\text{Hz}^{1/2}$): 243 K: 3.65, 253 K: 3.25, 263 K: 1.18, 273 K: 0.323, 283 K: 0.116, 293 K: 3.24×10^{-2} , 303 K: 1.45×10^{-2} ; stand. dev. of these values does not exceed 20%.

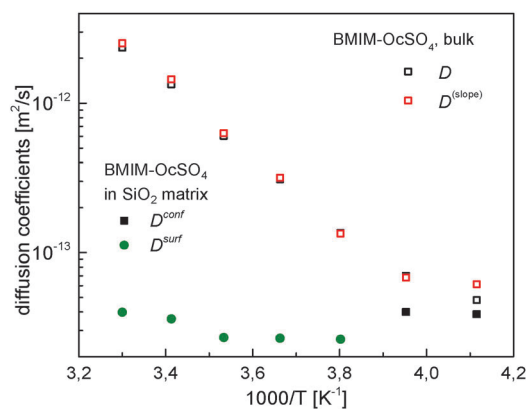


Fig. 6 Translational diffusion coefficients for BMIM-OcSO₄ in bulk and in a SiO₂ matrix (40 wt% of BMIM-OcSO₄).

This strategy cannot be continued for higher temperatures (263 K and above) when the differences in the shapes of the relaxation dispersion profiles as well as in the values of the relaxation rates for BMIM-OcSO₄ in bulk and in confinement become more significant. In this case we consider the overall relaxation rate, $R_1(\omega)$, as a sum of two contributions: $R_1(\omega) = R_1^{\text{core}}(\omega) + R_1^{\text{surf}}(\omega)$, resulting from a bulk-like fraction of the liquid near the center of the pore (a core fraction, $R_1^{\text{core}}(\omega)$) and from a fraction near the pore surface (a surface fraction, $R_1^{\text{surf}}(\omega)$). Anticipating the obtained result, it has turned out that the rotational dynamics of the cations and anions at 263 K and above can be treated as unaffected by the confinement. Nevertheless, as the analysis requires addition of a frequency independent term, C , we shall come back to this statement later.

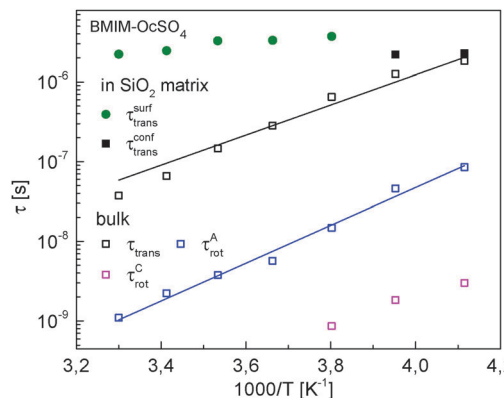


Fig. 7 Translational and rotational correlation times for BMIM-OcSO₄ in bulk and in an SiO₂ matrix (40 wt% of BMIM-OcSO₄); dashed lines – interpolations by means of the VFT (Vogel–Fulcher–Tammann) equation. The activation energies yield 36.2 kJ mol^{−1}, 45.4 kJ mol^{−1} and 49.6 kJ mol^{−1} for translational diffusion in bulk, anionic and cationic rotation in bulk, respectively.

Thus, the following expression has been applied to analyze the $R_1(\omega)$ data for the confined liquid at 263 K and above:

$$\begin{aligned}
 R_1(\omega) &= R_1^{\text{surf}}(\omega) + R_1^{\text{core}}(\omega) \\
 &= R_{1,\text{inter}}^{\text{surf}}(\omega) + R_{1,\text{inter}}^{\text{core}}(\omega) + R_{1,\text{intra}}^{\text{C,conf}} + R_{1,\text{intra}}^{\text{A,conf}} + C \\
 &= \frac{108}{5} \left(\frac{\mu_0}{4\pi} \gamma_{\text{H}}^2 \hbar \right)^2 \frac{N_{\text{H}}^{\text{surf}}}{d^3} \\
 &\quad \times \int_0^\infty \frac{u^2}{81 + 9u^2 - 2u^4 + u^6} \left(\frac{u^2}{u^4 + (\omega\tau_{\text{trans}}^{\text{surf}})^2} + 4 \frac{u^2}{u^4 + (2\omega\tau_{\text{trans}}^{\text{surf}})^2} \right) du \\
 &\quad + \frac{108}{5} \left(\frac{\mu_0}{4\pi} \gamma_{\text{H}}^2 \hbar \right)^2 \frac{N_{\text{H}}^{\text{core}}}{d^3} \\
 &\quad \times \int_0^\infty \frac{u^2}{81 + 9u^2 - 2u^4 + u^6} \left(\frac{u^2}{u^4 + (\omega\tau_{\text{trans}})^2} + 4 \frac{u^2}{u^4 + (2\omega\tau_{\text{trans}})^2} \right) du \\
 &\quad + C_{\text{DD}}^{\text{C}} \left(\frac{\tau_{\text{rot}}^{\text{C}}}{1 + (\omega\tau_{\text{rot}}^{\text{C}})^2} + 4 \frac{\tau_{\text{rot}}^{\text{C}}}{1 + (2\omega\tau_{\text{rot}}^{\text{C}})^2} \right) \\
 &\quad + C_{\text{DD}}^{\text{A}} \left(\frac{\tau_{\text{rot}}^{\text{A}}}{1 + (\omega\tau_{\text{rot}}^{\text{A}})^2} + 4 \frac{\tau_{\text{rot}}^{\text{A}}}{1 + (2\omega\tau_{\text{rot}}^{\text{A}})^2} \right) + C \quad (9)
 \end{aligned}$$

For clarity of the analysis it is worth discussing the individual terms of eqn (9) in more detail. The first term, $R_{1,\text{inter}}^{\text{surf}}(\omega)$, is associated with the translational dynamics near the pore surface, treated as a three-dimensional diffusion process (we shall comment on that later). It brings two adjustable parameters: the number of protons per unit volume in the vicinity of the surface, $N_{\text{H}}^{\text{surf}}$, and the translational correlation time near the surface, $\tau_{\text{trans}}^{\text{surf}}$; the distance of closest approach remains unchanged, $d = 4.2$ Å. The second term, $R_{1,\text{inter}}^{\text{core}}(\omega)$, corresponds to the translational dynamics near the pore center (core) and includes only one adjustable parameter that is the number of protons per unit volume in the core fraction, $N_{\text{H}}^{\text{core}}$ (one can expect that the number might be somewhat smaller than N_{H} as some of the ions are adsorbed on the surface). The translational correlation

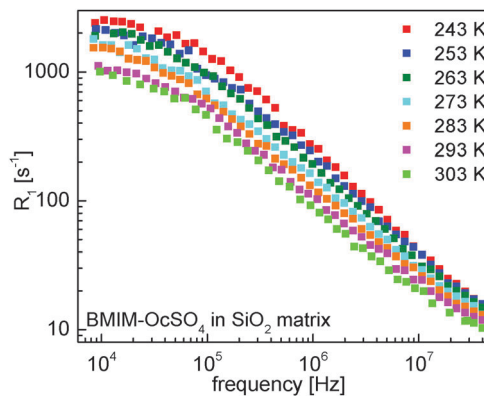


Fig. 8 ¹H spin–lattice relaxation dispersion profiles for BMIM-OcSO₄ in an SiO₂ matrix (40 wt% of BMIM-OcSO₄).

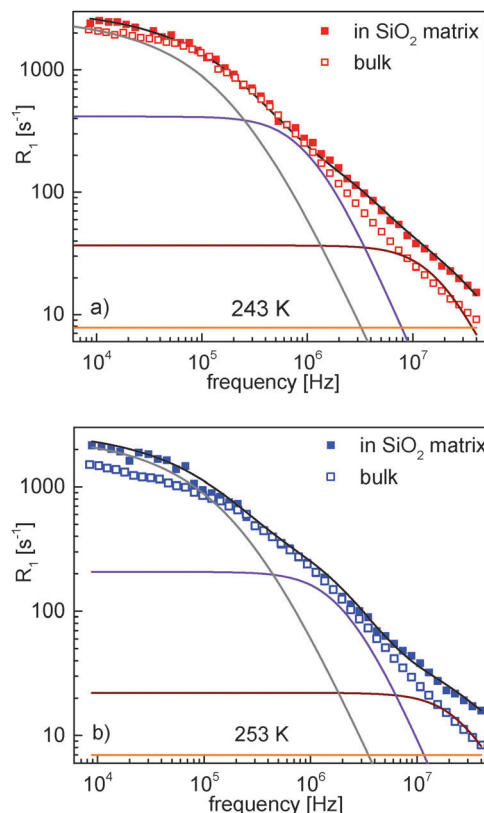


Fig. 9 (a and b) ¹H spin–lattice relaxation profiles for BMIM-OcSO₄ in an SiO₂ matrix (40 wt% of BMIM-OcSO₄) at (a) 243 K, (b) 253 K – solid squares, compared with the corresponding data for BMIM-OcSO₄ in bulk – open squares. Black lines – the overall relaxation rate ($R_1(\omega)$) decomposed into the contributions described by eqn (7): gray lines – inter-ionic (intermolecular) contribution ($R_{1,\text{inter}}(\omega)$), violet lines – anionic intramolecular contribution ($R_{1,\text{intra}}^{\text{A}}(\omega)$), brown lines – cationic intramolecular contribution ($R_{1,\text{intra}}^{\text{C}}(\omega)$), orange lines – a frequency independent contribution (C).

time in the core fraction is assumed to be the same as in bulk, τ_{trans} ; as the distance of closest approach also remains unchanged this implies the same diffusion coefficients in the core fraction and in bulk. The next two terms, $R_{1,\text{intra}}^{\text{C,conf}}(\omega)$ and $R_{1,\text{intra}}^{\text{A,conf}}(\omega)$, are associated with the rotational dynamics of the

Table 2 Parameters describing the translational and rotational dynamics of BMIM-OcSO₄ in an SiO₂ matrix ($d = 4.2$ Å, $C_{\text{DD}}^{\text{A}} = 9 \times 10^8$ Hz², $C_{\text{DD}}^{\text{C}} = 1.47 \times 10^9$ Hz², $N_{\text{H}} = 5.8 \times 10^{-2}$ Å⁻³)

T [K]	$\tau_{\text{trans}}^{\text{conf}}$ [s]	D^{conf} [m ² s ⁻¹]	$\tau_{\text{rot}}^{\text{A,conf}}$ [s]	$\tau_{\text{rot}}^{\text{C,conf}}$ [s]	C [s ⁻¹]	Stand. dev. (%)
243	2.28×10^{-6}	3.86×10^{-14}	9.25×10^{-8}	5.01×10^{-9}	7.8	43
253	2.21×10^{-6}	3.99×10^{-14}	4.60×10^{-8}	2.99×10^{-9}	7.0	49

cations and anions, respectively, in the confinement (not distinguishing between the core and surface fractions). As already anticipated, a comparison of the relaxation data for the bulk and confined liquid in the range of high frequencies has brought us to the conclusion that at higher temperatures the confinement does not cause significant changes in the rotational dynamics compared to bulk. As a consequence, the rotational correlation times (and the dipolar relaxation constants) have been set to the values obtained for the bulk liquid; this means that those terms do not bring additional fitting parameters. Eventually, the expression has been completed by adding a constant, C , which is needed to reproduce the relaxation data at high frequencies; this is the fourth adjustable parameter. This strategy applied to the data for 263 K and 273 K has led to good agreement with the experimental data (Fig. 10a and b) and the values of $\tau_{\text{trans}}^{\text{surf}}$ and C collected in Table 3; the calculated diffusion coefficients ($D^{\text{surf}} = d^2/(2\tau_{\text{trans}}^{\text{surf}})$) and $\tau_{\text{trans}}^{\text{surf}}$ values are shown in Fig. 6 and 7, respectively. The obtained numbers of protons per unit volume yield: $N_{\text{H}}^{\text{core}} = 5.22 \times 10^{-2}$ Å⁻³, $N_{\text{H}}^{\text{surf}} = 2.7 \times 10^{-2}$ Å⁻³. Fig. 10a and b shows the overall, theoretical relaxation dispersion profile, $R_1(\omega)$, decomposed into the individual contributions: $R_{1,\text{inter}}^{\text{surf}}(\omega)$, $R_{1,\text{inter}}^{\text{core}}(\omega)$, $R_{1,\text{intra}}^{\text{C,conf}}(\omega)$, $R_{1,\text{intra}}^{\text{A,conf}}(\omega)$ and C . As already explained, the intramolecular relaxation contributions are the same as in bulk: $R_{1,\text{intra}}^{\text{A}}(\omega) = R_{1,\text{intra}}^{\text{C}}(\omega)$ and $R_{1,\text{intra}}^{\text{C,conf}}(\omega) = R_{1,\text{intra}}^{\text{C}}(\omega) = 5C_{\text{DD}}^{\text{C}}\tau_{\text{rot}}^{\text{C}}$ (for 263 K and 273 K the $R_{1,\text{intra}}^{\text{C}}(\omega)$ term is already frequency independent and yields: 263 K: 6.31 s⁻¹; 273 K: 3.66 s⁻¹). The same strategy has been applied to the relaxation data at 283 K, 293 K and 303 K leading to a good agreement with the experimental results as shown in Fig. 11, and consistent parameters. For these temperatures the overall relaxation rates, $R_1(\omega)$, have been decomposed in Fig. 11 into the $R_{1,\text{inter}}^{\text{surf}}(\omega)$ contribution and a sum of the other terms present in eqn (9). The obtained parameters are included into Table 3; $\tau_{\text{trans}}^{\text{surf}}$ and the calculated D^{surf} are shown in Fig. 7 and 6, respectively. Fig. 12 shows the differences between the experimental values of the relaxation rate $R_1(\omega)$ and the theoretical term: $\tilde{R}_1(\omega) = R_{1,\text{inter}}^{\text{core}}(\omega) + R_{1,\text{intra}}^{\text{C,conf}}(\omega) + R_{1,\text{intra}}^{\text{A,conf}}(\omega) + C$, plotted versus $\sqrt{\omega}$, for the temperatures 283 K, 293 K and 303 K (the results for 263 K and 273 K are not shown as they are similar to those for 283 K). Although the linear dependencies are not surprising taking into account that the $R_{1,\text{inter}}^{\text{surf}}(\omega)$ term well describes the low frequency part of the relaxation dispersion profiles for the confined liquid, they confirm the assumption that this part of the profiles reflects a three dimensional diffusion process.

5. Discussion

Let us begin the discussion with BMIM-OcSO₄ in bulk. In this case the ¹H spin-lattice relaxation dispersion profiles have been decomposed into three contributions: a term associated

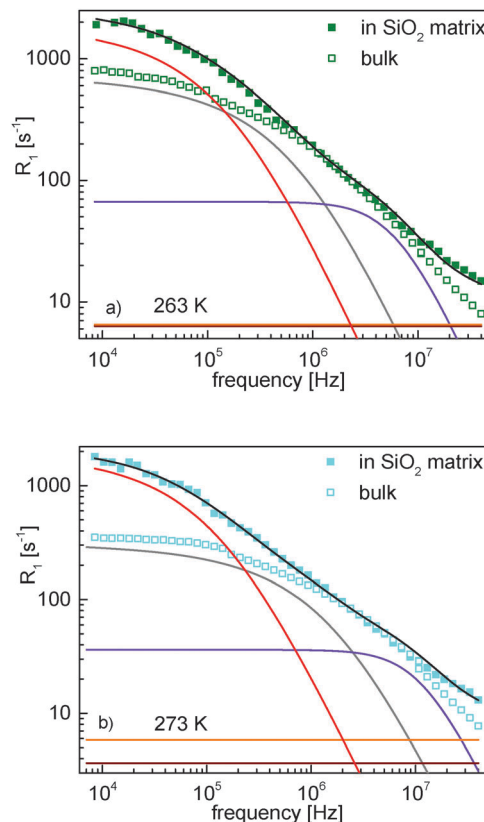


Fig. 10 (a and b) ¹H spin-lattice relaxation profiles for BMIM-OcSO₄ in an SiO₂ matrix (40 wt% of BMIM-OcSO₄) at (a) 263 K, (b) 273 K – solid squares, compared with the corresponding data for BMIM-OcSO₄ in bulk – open squares. Black lines – the overall relaxation rate, $R_1(\omega)$, decomposed into the contributions described by eqn (8): red lines – $R_{1,\text{inter}}^{\text{surf}}(\omega)$, gray lines – $R_{1,\text{inter}}^{\text{core}}(\omega)$, violet lines – $R_{1,\text{intra}}^{\text{A,conf}}(\omega) = R_{1,\text{intra}}^{\text{A}}(\omega)$, brown lines – $R_{1,\text{intra}}^{\text{C,conf}}(\omega) = R_{1,\text{intra}}^{\text{C}}(\omega) = 5C_{\text{DD}}^{\text{C}}\tau_{\text{rot}}^{\text{C}}$, orange lines – a frequency independent contribution, C .

Table 3 Parameters describing the translational and rotational dynamics of BMIM-OcSO₄ in the SiO₂ matrix ($d = 4.2$ Å, $C_{\text{DD}}^{\text{A}} = 9 \times 10^8$ Hz², $C_{\text{DD}}^{\text{C}} = 1.47 \times 10^9$ Hz², $N_{\text{H}}^{\text{core}} = 5.22 \times 10^{-2}$ Å⁻³, $N_{\text{H}}^{\text{surf}} = 2.7 \times 10^{-2}$ Å⁻³)

T [K]	$\tau_{\text{trans}}^{\text{surf}}$ [s]	D^{surf} [m ² s ⁻¹]	C [s ⁻¹]	Stand. dev. (%)
263	3.37×10^{-6}	2.62×10^{-14}	6.52	35
273	3.32×10^{-6}	2.66×10^{-14}	5.84	30
283	3.28×10^{-6}	2.69×10^{-14}	5.53	11.6
293	2.46×10^{-6}	3.59×10^{-14}	5.07	18.3
303	2.22×10^{-6}	3.98×10^{-14}	4.50	21.8

with the translational dynamics of both ions (it has been assumed that translational diffusion coefficients of the cation and the anion are equal) and two terms associated with the rotational motion of the anion and the cation, respectively. The assumption that the translational motion of both ions might be described by one diffusion coefficient can be considered as problematic although it has been shown in the literature^{21,32,33} that the diffusion of both ions in ionic liquids is pretty similar – one might find this argument not fully convincing, anyway. One can, however, also look at this subject from a different side: the relaxation data confirm this assumption – if the diffusion coefficients would differ much, we would be not able to get such

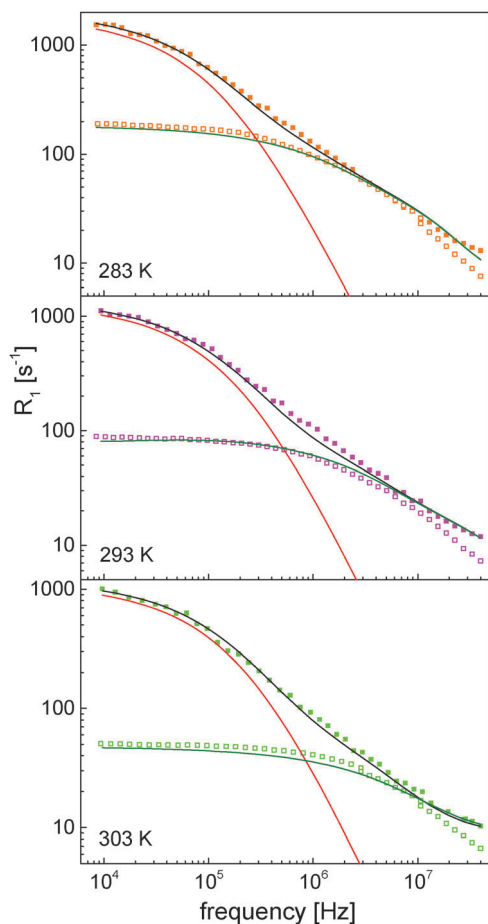


Fig. 11 ^1H spin-lattice relaxation profiles of BMIM-OcSO₄ in an SiO₂ matrix (40 wt% of BMIM-OcSO₄) at 283 K, 293 K and 303 K – solid squares, compared with the corresponding data for BMIM-OcSO₄ in bulk – open squares. Black lines – the overall relaxation rate, $R_1(\omega)$, decomposed into $R_{1,\text{intra}}^{\text{surf}}(\omega)$ – red lines and a sum of the other terms of eqn (9) – green lines.

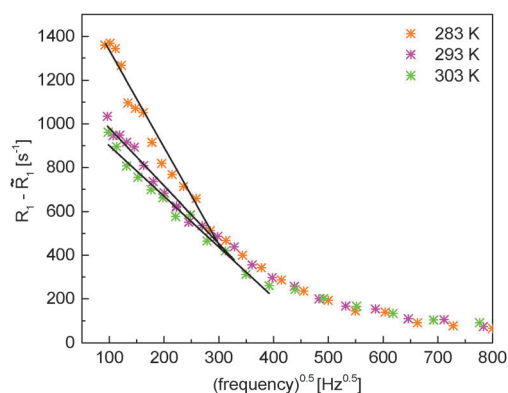


Fig. 12 Linear dependencies of the difference between the experimental values of $R_1(\omega)$ and the theoretically calculated term: $\tilde{R}_1(\omega) = R_{1,\text{intra}}^{\text{core}}(\omega) + R_{1,\text{intra}}^{\text{conf}}(\omega) + R_{1,\text{intra}}^{\text{surf}}(\omega) + C$ versus square root of frequency, $\sqrt{\omega}$.

good agreement with the experimental data. The translational correlation time, τ_{trans} , fulfils the Vogel-Fulcher-Tammann (VFT) equation³⁴ with an activation energy of 36.2 kJ mol⁻¹.

The translational diffusion coefficient changes in the temperature range of 243–303 K by a factor of about 500: from about $5 \times 10^{-14} \text{ m}^2 \text{ s}^{-1}$ to about $2.5 \times 10^{-12} \text{ m}^2 \text{ s}^{-1}$. The rotational correlation time of the anion, $\tau_{\text{rot}}^{\text{A}}$, varies between about $8.5 \times 10^{-8} \text{ s}$ and $2 \times 10^{-9} \text{ s}$, while the rotational correlation time of the cation, $\tau_{\text{rot}}^{\text{C}}$, is shorter by a factor of about 30 at 243 K ($3.0 \times 10^{-9} \text{ s}$) and a factor of about 80 at 296 K ($2.5 \times 10^{-11} \text{ s}$); the correlation times cannot be defined at 303 K because they are too short. Taking into account the viscosity of BMIM-OcSO₄ that yields 0.87 Pa s at 293 K,³⁵ one obtains from the Stokes-Einstein equation about 2 Å for the radius. This value is somewhat too small, but reasonable taking into account the approximated character of the Stokes-Einstein equation. Both rotational correlation times fulfill the VFT equation with the activation energies of 45.4 kJ mol⁻¹ and 49.6 kJ mol⁻¹, for the anion and the cation, respectively. It is of interest to consider the ratios $\tau_{\text{trans}}/\tau_{\text{rot}}^{\text{A}}$ and $\tau_{\text{trans}}/\tau_{\text{rot}}^{\text{C}}$. As already explained, according to the Stokes-Einstein equation for spherical molecules in the hydrodynamic regime, the ratio $\tau_{\text{trans}}/\tau_{\text{rot}}$ yields 9.¹¹ It has, however, been shown that for “real” molecular liquids the ratio yields a larger value: 30–50.²⁶ To our knowledge this issue has not been discussed so far for ionic liquids. For BMIM-OcSO₄ the ratio $\tau_{\text{trans}}/\tau_{\text{rot}}^{\text{A}}$ varies between about 22 (243 K) and 32 (296 K) similar to several molecular liquids, while for $\tau_{\text{rot}}^{\text{A}}/\tau_{\text{rot}}^{\text{C}}$ one gets 28 (243 K) and 82 (296 K) – the rotation of the cation is much faster than that of the anion as the former is considerably smaller.

The ^1H spin-lattice relaxation data for BMIM-OcSO₄ in the SiO₂ matrix indicate a considerable influence of the confinement on the ionic dynamics. Nevertheless, at the lowest temperatures, 243 K and 253 K, the effects at low frequencies, where the relaxation rate is dominated by translational dynamics, are relatively small and they can be explained by a translational motion somewhat slower than in bulk; the diffusion coefficients differ by a factor of about 1.5 from the values for the bulk. The influence of confinement on the rotational dynamics at those temperatures can also be explained by relatively small changes in the rotational correlation times: $\tau_{\text{rot}}^{\text{C}}$ in confinement increases by a factor of about 1.7 compared to the values for the bulk, while $\tau_{\text{rot}}^{\text{A}}$ remains (almost) unchanged. However, an additional motional process (not observed in bulk) has been revealed in confinement. The correlation time of this process is of the order of 10^{-9} s or shorter – therefore the corresponding relaxation contribution is frequency independent (represented as a constant term, C). This process can be attributed to the dynamics of the long chain of OcSO₄ anions. This motion slows down in confinement compared to bulk and therefore it becomes detectable in the relaxation experiment although still only as a constant. Actually, at higher temperatures (263 K and above) the presence of this process is sufficient to explain the differences between the relaxation rates for bulk and confinement observed at higher frequencies; at 263 K and above the rotational correlation times ($\tau_{\text{rot}}^{\text{A}}$ and $\tau_{\text{rot}}^{\text{C}}$) remain unaffected by the confinement. However, one should stress that at 253 K and 243 K, the rotational dynamics in confinement begin to slow down – it starts from the cations at 253 K: $\tau_{\text{rot}}^{\text{C,conf}}(253 \text{ K}) = 2.99 \times 10^{-9} \text{ s}$, while for bulk $1.83 \times 10^{-9} \text{ s}$; then at 243 K the effect becomes

more pronounced and it is observed for both ions: $\tau_{\text{rot}}^{\text{C,conf}} = 5.01 \times 10^{-9}$ s (bulk – 3.00×10^{-9} s), $\tau_{\text{rot}}^{\text{A,conf}} = 9.25 \times 10^{-8}$ s (bulk – 8.49×10^{-8} s). The trend indicates that with a further decrease in temperature a progressively more significant slowing down of the rotational dynamics in confinement will occur. Similar observations have been described in the literature. In ref. 36 a very small slowing-down of the rotational dynamics of BMIM-TFSI confined in silica matrices has been reported. The system is highly similar to this one discussed in our manuscript. The applied method – ^1H NMR spectroscopy performed at high frequency (400 MHz) is a very sensitive method for probing rotational dynamics with atomistic resolution. In this paper spin-lattice relaxation times for ^1H nuclei at different positions in the cation are compared with the relaxation times in confinement. The differences in the relaxation times, and hence the rotational dynamics, for the cases of bulk and confinement are small. The dynamics of ammonium based ionic liquids in silica nanopores have recently been studied³⁷ by means of NMR spectroscopy. Also in this case it has been concluded that the rotational dynamics only marginally slows down in confinement. Actually liquid-like behavior of confined liquids below their crystallization temperatures has been reported; for instance water confined in nanotubes shows a liquid-like behavior at temperatures as low as about 50 K.³⁸ Nevertheless, there are also systems that show a much more significant slowing down of the rotational dynamics in confinement.^{39,40}

The divergences between the ^1H spin-lattice relaxation rates at low frequencies for BMIM-OcSO₄ in confinement and in bulk become progressively more significant with increasing temperature reaching a ratio about 20 at 303 K. So a large effect cannot be described by a slight slowing down of the translational diffusion. Therefore the model assumes that the relaxation rates are a sum of two terms: a contribution originating from a bulk-like (core) fraction described by the parameters obtained from the analysis of the relaxation data for bulk, and a contribution originating from a much slower diffusing fraction of ions near the surface. It has turned out that the diffusion coefficient near the surface, D^{surf} , is only weakly dependent on temperature and the ratio D/D^{surf} (D denotes the diffusion coefficient in bulk) varies from about 5 at 263 K to 60 at 303 K. In this context the diffusion coefficient D^{conf} obtained for 243 K and 253 K can be treated as an effective parameter accounting for the small slowing down of the translational motion near the surface at the lowest temperatures compared to the core fraction. When the distribution of the translational correlation time becomes broader (with increasing temperature) one cannot reproduce the relaxation data in terms of a single, effective correlation time. In this context, the concept of the two contributions to the relaxation associated with the core and the surface fractions can be treated as a bi-modal distribution of the translational correlation times. The assumption of such a distribution is a simplification of the problem, however, it reflects the essential effects of the confinement on the translational dynamics. The parameters $N_{\text{H}}^{\text{core}} = 5.22 \times 10^{-2} \text{ \AA}^{-3}$ and $N_{\text{H}}^{\text{surf}} = 2.69 \times 10^{-2} \text{ \AA}^{-3}$ can be treated as an indication of the ratio between the fraction of ions diffusing freely and the

fraction of ions for the moment adsorbed to the wall. The effect of the influence of the confinement increasing with increasing temperature can be explained as follows: when the translational dynamics is slow (low temperatures) the fact that some of the ions remain for a time comparable to the translational correlation time attached to the surface of the pores does not change the effective translational correlation time significantly. Nevertheless, with increasing temperature the diffusion in the core (bulk-like) fraction becomes faster, while the ions near the surface still remain attached to it for a quite long time, the translational dynamics effectively slows down significantly. This argument can be supported by another one: if the residence lifetime of the ions on the surface would be comparable (or shorter) than the translational dynamics in bulk, the attached ions would fast exchange with the free ones and, in the end effect, the translational motion of all ions could be described by a single correlation time as it has been possible for 243 K and 253 K, but not for higher temperatures. Inspecting the translational correlation times obtained for the liquid in bulk and in confinement near the surface one can estimate the residence lifetime of the ions on the surface as being of the order of 10^{-6} s. The translational correlation time near the surface is of the order of 10^{-6} s even at 303 K, at which the corresponding correlation time for bulk reaches a value of the order of 10^{-8} s which indicates that the motion near the surface is controlled by a long residence lifetime. One can also conclude that for BMIM-OcSO₄ in the SiO₂ matrix we observe the weak adsorption case,²² *i.e.* the ions escape to the core fraction after a single adsorption act. This conclusion is based on the fact that it has turned out to be possible to reproduce the relaxation data at low frequencies (for the confined liquid) in terms of three dimensional diffusion. Moreover, Fig. 12 explicitly shows this effect – the contribution to the overall relaxation associated with the diffusion near the surface ($R_1(\omega) - \tilde{R}_1(\omega)$) shows a linear dependence on $\sqrt{\omega}$, characteristic of three dimensional diffusion. For the strong adsorption case, when ions leave the surface after many adsorption/desorption acts, the diffusion process would rather show a two dimensional character.

The mechanism of the interactions of cations and anions with the pore surface walls is described in detail in ref. 41.

6. Conclusions

Analysis of the extensive ^1H spin-lattice relaxation data set for BMIM-OcSO₄ in bulk and in SiO₂ confinement has led to several conclusions and observations. We summarize them below. The data for BMIM-OcSO₄ in bulk show that both ions exhibit very similar translational dynamics – it can be described by one diffusion coefficient significantly varying in the temperature range of 243–303 K from about $5 \times 10^{-14} \text{ m}^2 \text{ s}^{-1}$ to about $2.5 \times 10^{-12} \text{ m}^2 \text{ s}^{-1}$. The data allow for separating the rotational dynamics of the anion and the cation. The rotational correlation time of the anion, $\tau_{\text{rot}}^{\text{A}}$, varies between about 8.5×10^{-8} s and 2×10^{-9} s that gives a factor of about 400–500 ($\tau_{\text{rot}}^{\text{A}}$ at 303 K cannot be determined), similar to the diffusion coefficient.

The ratio $\tau_{\text{trans}}/\tau_{\text{rot}}^{\text{A}}$ yields a value of 20–30 typical of molecular liquids. The rotational correlation time of the cation, $\tau_{\text{rot}}^{\text{C}}$, is by at least one order of magnitude shorter than $\tau_{\text{rot}}^{\text{A}}$. All processes follow the VFT equation with activation energies of the order of 45–50 kJ mol⁻¹. Confinement, in fact, does not affect the rotational dynamics (except for small effects at the lowest temperature, 243 K). It reveals, however, an additional motional process not observed in bulk that manifests itself as a frequency independent term indicating that the correlation time of this process is on the ns or an even shorter scale. In contrast to the rotational motion, the translational dynamics is strongly affected by the confinement. Assuming that the relaxation contribution associated with the translational dynamics results from two components: a term representing translation near the pore center (core-fraction) and a term representing translation near the pore surface mediated by acts of adsorption to the pore walls, it has been shown that the effective movement near the surface is described by a correlation time $\tau_{\text{trans}}^{\text{surf}}$ of the order of 10⁻⁶ s (3.37×10^{-6} at 263 K and 2.22×10^{-6} at 303 K). This indicates that the translational motion is considerably slowed down by the adsorption acts.

In the final concluding sentence we wish to stress the virtue of NMR relaxometry that gives deep insight into molecular (ionic) dynamics as demonstrated by the example discussed in this work.

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